

1. Administrative & Study Identification

Full Study Title: An International, Prospective, Open-label, Multi-center, Randomized Phase III Study Comparing Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) Versus Observation to Delay Castration or Disease Recurrence in Adult Male Patients With Prostate-specific Membrane Antigen (PSMA) Positive Oligometastatic Prostate Cancer (OMPC) **Short Title/Acronym:** PSMA-DC **Protocol Number:** CAAA617D12302 **NCT Number:** NCT05939414 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Lutetium (177Lu) vipivotide tetraxetan (AAA617) and Stereotactic Body Radiation Therapy (SBRT) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of lutetium (177Lu) vipivotide tetraxetan versus observation (post-SBRT) in delaying metastasis or death in patients with PSMA-positive oligometastatic prostate cancer. **Secondary Objectives:** To evaluate Time to Hormonal Therapy (TTHT), Time to prostate-specific antigen (PSA) progression (TTPSAP), Radiographic Progression Free Survival (rPFS), Time to next therapy (local or systemic), Time to symptomatic progression, Overall Survival (OS), and Patient-reported health-related quality of life.

2.2 Background & Rationale To provide evidence for the treatment of AAA617 in early-stage prostate cancer patients to control recurrent tumors from progressing to fatal metastatic disease. This systemic intensification approach aims to preserve the patient's quality of life by delaying the initiation and toxicities of systemic androgen deprivation therapy (ADT).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Observation (SBRT alone vs. SBRT + AAA617) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 450 male participants **Number of Sites:** Multi-center (International) **Overall Status:** Recruiting **Estimated Study Start:** 12-Mar-2024 **Estimated Primary Completion:** 03-Jun-2030 **Estimated Study Completion:** Oct-2031

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult males 18 to 100 years of age. 2. Histologically confirmed prostate cancer with biochemically recurrent disease after definitive treatment (Radical Prostatectomy or External Beam Radiation Therapy). 3. Oligometastatic Prostate Cancer (OMPC) with 1 to 5 PSMA-positive metastatic lesions (at least 1 distant M1 lesion) on a screening PSMA PET/CT scan. 4. Negative conventional imaging (CT/MRI and bone scans) for M1 disease at screening.

4.2 Exclusion Criteria 1. Participants with *de novo* (synchronous) OMPC at screening. 2. Prior therapy with ADT and Androgen Receptor Pathway Inhibitors (ARPIs) used for metastatic prostate cancer treatment. 3. Unmanageable concurrent bladder outflow obstruction or urinary incontinence.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Stereotactic Body Radiation Therapy (SBRT) administered to all metastatic lesions (~3 weeks duration). Participants in the investigational arm will then receive Lutetium (177Lu) vipivotide tetraxetan (AAA617) at 7.4 GBq every 6 weeks for up to 4 cycles. **Route of Administration:** External beam (SBRT) and Intravenous injection (177Lu-PSMA-617). **Duration of Treatment:** SBRT (~3 weeks) + 4 cycles of AAA617 (~24 weeks). Total study observation duration is approximately 6.5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care treatments for manageable bladder outflow obstruction/incontinence. **Prohibited:** Concurrent Androgen Deprivation Therapy (ADT), ARPIs, immunotherapy, systemic chemotherapy, and other radiopharmaceutical agents (e.g., Strontium-89).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, PSMA PET/CT Scan (68Ga or 18F), Baseline CI
Baseline	Day 0	Randomization, SBRT Initiation
Interim	Every 6 Weeks	AAA617 Administration (up to 4 cycles), Safety Labs, Efficacy Assessments
End of Study	Up to 6.5 Years	Long-term follow-up, Metastasis Free Survival assessment, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Metastasis Free Survival (MFS) - defined as the time from randomization until the first evidence of radiographically detectable bone or soft tissue distant

metastasis or death due to any cause. **Measurement Method:** Assessed by a Blinded Independent Review Committee (BIRC) via conventional imaging (CT/MRI and bone scans) up to approximately 30 months.

7.2 Secondary & Exploratory Endpoints * Time to Hormonal Therapy (TTHT) * Radiographic Progression Free Survival (rPFS) * Time to prostate-specific antigen (PSA) progression (TTPSAP) * Time to next therapy (local or systemic) * Time to symptomatic progression * Overall Survival (OS) * Patient-reported health-related quality of life (FACT-P, FACT-RNT, BPI-SF)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection evaluating hematological toxicities (e.g., anemia, thrombocytopenia) and radioligand-specific events (e.g., dry mouth, fatigue) via regular oncology assessments. **Serious Adverse Events (SAEs):** Monitored throughout the study duration and reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms implemented by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary MFS outcomes. Safety Set for AE/SAE evaluations. **Sample Size Justification:** Target $N = 450$ appropriately powered to detect statistically significant delays in MFS and time to ADT initiation. **Handling of Missing Data:** Standard right-censoring mechanisms will be applied for survival endpoints (MFS, rPFS, OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits by the sponsor to ensure protocol adherence. **Audit Trail:** Detailed eCRF locking, tracking of treatment cycles, and independent BIRC assessments of all imaging data.

11. Study Identifiers & Governance

Primary ID: CAAA617D12302 **Registry Number:** NCT05939414 **Other Identifier:** 2022-502956-29-00 (EU CT Number) **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** PSMA-DC **Official Regulatory Title:** An Open-label Study Comparing Lutetium

12. Clinical Framework (Oncology Specific)

Primary Condition: Oligometastatic Prostate Cancer (OMPC)
Diagnosis Threshold: Biochemically recurrent disease with 1 to 5 PSMA-positive lesions **Medical Methodology:** Metastasis-directed therapy (SBRT) combined with systemic Radioligand Therapy
Optimization Target: Delay of Androgen Deprivation Therapy (ADT) initiation and metastasis prevention

13. Study Procedures & Methodology

Management Pathway: SBRT followed by up to 4 cycles of 177Lu-PSMA-617 **Education Protocol (HCP):** Administration protocol and radiation safety training for handling 177Lu **Education Protocol (Patient):** Radiation precautions post-infusion and tracking of radioligand side effects **Quality Audit Mechanism:** BIRC centralized review of imaging (CT/MRI/Bone scans) to ensure unbiased outcome measurements

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6.5 years **Onsite Visit Cadence:** Every 6 weeks during the treatment phase (Cycles 1-4) **Remote Monitoring Frequency:** Regular interval follow-up post-treatment for survival and disease progression metrics **Checklist Review Interval:** Continuous assessment up to radiographic progression or initiation of next-line therapy

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				